

Master 2 Research Report

LLM-Enhanced Graph-Based Intrusion Detection

A Parallel GNN–LLM Framework with Bidirectional
Refinement

Ali Alzein

Supervised by Fouad Al Tfaily

CESI LINEACT / Lebanese University

Contents

1	Introduction and Context	2
1.1	Why combine graph and language representations?	2
1.2	Research problem and questions	4
1.3	Objectives and contributions	4
1.4	Report structure	5

1 Introduction and Context

Intrusion detection is an operational problem as well as a classification problem. Missed attacks can leave harmful activity unchallenged, while false alarms consume analyst attention and make genuine incidents harder to prioritise. Practical guidance therefore treats alert validation, false-positive identification, and threat prioritisation as central parts of intrusion monitoring [6].

Modern networks also produce a continuous stream of connections whose meaning is rarely contained in a single record. A destination port, a packet count, or a flow duration can be informative, but the surrounding communication pattern often matters just as much. A distributed denial-of-service attack may appear as many sources converging on one destination, whereas a scan may appear as one source contacting many hosts or services. This relational character makes network traffic a natural candidate for graph-based analysis. At the same time, network events also carry semantic cues: protocol and port combinations, traffic magnitudes, and recurring behavioural descriptions can suggest what an interaction represents. An intrusion detection system that sees only one of these views is therefore working with an incomplete account of the traffic.

This report investigates whether structural and semantic evidence can be combined in a single edge-level intrusion detection framework using NF-UNSW-NB15. One experimental boundary must be stated from the outset: the current graph construction aggregates flows by source IP, destination IP, and attack class. Ground-truth labels therefore participate in defining edge identity, even though the attack name is excluded from the model’s input features and generated text. The consequence is explicit: the resulting scores compare representations and interaction mechanisms under a controlled protocol, but they are not deployment-valid estimates of detection performance on unseen, unlabelled traffic.

The original UNSW-NB15 benchmark was created from a mixture of normal activity and contemporary attacks in the UNSW Canberra Cyber Range Laboratory [4]. Its NetFlow derivative expresses that traffic through a compact, practically relevant feature set intended to support reproducible intrusion-detection research [5]. This study uses that dataset to examine what each branch learns, whether their combination is useful, and whether semantic feedback improves the graph representation rather than merely changing the final classifier.

1.1 Why combine graph and language representations?

Graph neural networks (GNNs) provide a direct way to represent communication structure. In this work, IP addresses form the nodes of a directed graph and aggregated flows

form labelled edges. A graph-attention encoder propagates information between communicating hosts and produces a structural representation for each edge. This branch can learn patterns that are difficult to express as independent rows in a conventional feature table. It does not, however, possess an explicit vocabulary for interpreting the traffic. The graph tells the model *who communicated with whom* and how that interaction is positioned in the network; it does not automatically explain what a high-frequency TCP connection to a particular service may signify.

A pretrained cybersecurity language encoder offers a complementary view. The same aggregated flow can be written as a short, controlled sentence describing its endpoints, protocol, port, frequency, volume, and duration. A language encoder can then map that description into a semantic embedding. The implementation uses CySecBERT, a BERT model adapted to cybersecurity text and evaluated by its authors on domain-specific representation and classification tasks [1]. Throughout this report, this component is called the *semantic branch*. The deployed encoder is a BERT-class pretrained language model rather than a generative large language model; the more precise term prevents the project shorthand “LLM” from overstating its role.

Neither representation is assumed to be superior in every case. The central idea is instead to let them challenge and complement one another. Phase 1 runs the graph and semantic encoders in parallel and combines their edge embeddings through Adaptive Gated Attention Fusion (AGAF). Phase 2 adds uncertainty-guided semantic feedback: edges with high-entropy GNN predictions are selected for semantic consultation, and the resulting signal is reintroduced as a bias in the next graph-attention update. In intuitive terms, the graph branch indicates where it is uncertain, while the semantic branch supplies an additional judgement for those cases.

This general dual-stream idea is not novel by itself. Parallel BERT and graph encoders with attention-based fusion have already been studied outside network security, for example in misinformation detection [3]. Work in network intrusion detection has also combined GNNs and generative language models, but sometimes for a different purpose. XG-NID, for instance, performs intrusion classification with a heterogeneous GNN and uses an LLM after classification to generate explanations and remedial guidance [2]. The gap addressed here is narrower and more testable: a shared NetFlow pipeline in which a graph encoder and a cybersecurity language encoder both contribute directly to edge classification, followed by an explicit test of whether iterative semantic feedback improves the classifier beyond late fusion alone.

1.2 Research problem and questions

The research problem is not simply whether adding a language embedding raises a score. If one branch is already a strong classifier, a fusion module can appear successful while doing little more than copying that branch. Likewise, a feedback loop can improve the final output because semantic features are fused at the classifier, even when the claimed attention re-entry mechanism contributes nothing. The evaluation must therefore separate the value of the two modalities, the fusion mechanism, the uncertainty-selection policy, and the iterative update itself.

The study is organised around four research questions:

- RQ1.** How do the structural GNN branch and the semantic branch perform individually on NF-UNSW-NB15 under the same five-fold out-of-fold evaluation protocol?
- RQ2.** Does AGAF outperform both single-branch baselines on NF-UNSW-NB15, and are the observed differences supported by uncertainty estimates and robust comparisons?
- RQ3.** Does uncertainty-guided semantic feedback add value beyond static fusion? If so, does the gain come from semantic re-entry into graph attention or from final-output fusion?
- RQ4.** How do graph topology, repeated endpoint pairs, class imbalance, and the strength of the semantic consultant affect fusion and feedback on NF-UNSW-NB15?

These questions deliberately allow a negative answer. The purpose of the experiments is to determine when the proposed interaction works, when it does not, and why. This matters because a model ladder—GNN, semantic encoder, fusion, then feedback—is only meaningful if each higher rung is compared fairly with the strongest relevant lower rung.

1.3 Objectives and contributions

The practical objective is to build one reproducible pipeline that transforms NF-UNSW-NB15 NetFlow records into two aligned views and evaluates every stage under one consistent protocol. The work makes four contributions.

1. It implements a directed, edge-classification graph pipeline in which flows are aggregated by source IP, destination IP, and attack class, with centrality-based node features and statistical edge attributes.

2. It constructs a corresponding internal knowledge representation and serialises each edge as controlled natural language. The text supplied to the encoder is kept label-free by an explicit validation guard, preventing the semantic branch from seeing the target attack name or a synonymous relation label.
3. It implements and compares two multimodal stages: AGAF for parallel structural-semantic fusion, and uncertainty-guided semantic feedback that injects semantic evidence into graph attention for uncertain edges.
4. It provides a controlled evaluation on NF-UNSW-NB15, including unimodal baselines, fusion comparisons, uncertainty-selection ablations, trained-head baselines, and mechanism-isolation tests. It reports both positive and negative results to show which components carry the measured gain.

The contribution is therefore not presented as a universal improvement over existing intrusion detectors. It is an implementation and controlled evaluation of a particular form of structural-semantic cooperation, together with evidence about its limits. Deployment-valid evaluation remains open and requires graph construction in which edge identity can be determined without access to the target class.

1.4 Report structure

The remainder of the report follows four stages. The next section reviews graph-based intrusion detection, cybersecurity language encoders, and the main ways graph and language representations have been combined. The methodology section then describes the datasets, graph and text construction, the two encoders, AGAF, semantic feedback, and the evaluation protocol in enough detail to reproduce the experiments. The results section presents the NF-UNSW-NB15 performance ladder, statistical comparisons, ablations, and verified negative results. The final section interprets those findings, distinguishes supported conclusions from open claims, and identifies the experiments needed to move from an offline research prototype toward a deployment-valid intrusion detection system.

References

- [1] Markus Bayer, Philipp Kuehn, Ramin Shanehsaz, and Christian Reuter. Cy-SecBERT: A domain-adapted language model for the cybersecurity domain, 2022. arXiv:2212.02974.
- [2] Yasir Ali Farrukh, Syed Wali, Irfan Khan, and Nathaniel D. Bastian. XG-NID: Dual-modality network intrusion detection using a heterogeneous graph neural network and large language model, 2024. arXiv:2408.16021.
- [3] Hejamadi Rama Moorthy, N. J. Avinash, N. S. Krishnaraj Rao, K. R. Raghunandan, Radhakrishna Dodmane, Jeremy Joseph Blum, and Lubna A. Gabralla. Dual stream graph augmented transformer model integrating BERT and GNNs for context aware fake news detection. *Scientific Reports*, 15:25436, 2025. doi: 10.1038/s41598-025-05586-w.
- [4] Nour Moustafa and Jill Slay. UNSW-NB15: A comprehensive data set for network intrusion detection systems. In *2015 Military Communications and Information Systems Conference (MilCIS)*, pages 1–6, 2015. doi: 10.1109/MilCIS.2015.7348942.
- [5] Mohanad Sarhan, Siamak Layeghy, Nour Moustafa, and Marius Portmann. Netflow datasets for machine learning-based network intrusion detection systems. In *Big Data Technologies and Applications*, pages 117–135. Springer International Publishing, 2021. doi: 10.1007/978-3-030-72802-1_9.
- [6] Karen Scarfone and Peter Mell. Guide to intrusion detection and prevention systems (IDPS). Technical Report NIST Special Publication 800-94, National Institute of Standards and Technology, 2007.